

# PROJECT TECHNICAL REPORT

## Audio Signal Analysis for Emotion Recognition

### 1. Executive Summary

This technical report outlines the development of a real-time system designed to classify human emotional states through vocal signal intensity. The system utilizes an integrated hardware-software framework to capture, process, and visualize acoustic data for immediate emotional insight.

### 2. System Architecture

The system is architected into two fundamental operational layers:

- **Hardware Layer:** Centered around an [Arduino Uno](#) microcontroller and a high-sensitivity analog microphone sensor for raw data acquisition.
- **Software Layer:** A [Python-based](#) processing engine that manages high-speed serial communication, signal filtering, and the graphical user interface.

### 3. Hardware Implementation Logic

The hardware is optimized for high-fidelity signal capture using a **Peak-to-Peak** voltage measurement technique:

- **Acoustic Sampling:** The sensor captures sound pressure variations over a defined 50ms window.
- **Amplitude Calculation:** By isolating the maximum and minimum voltage peaks within the window, the system calculates the true signal amplitude, effectively neutralizing DC offset.

- **Data Transmission:** Processed intensity values are streamed to the software environment via serial interface at 9600 baud.

## 4. Software Processing & DSP

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The software backend employs Digital Signal Processing (DSP) to ensure classification accuracy:

- **Adaptive Baseline Calibration:** The system identifies the ambient noise floor at startup, creating a dynamic reference point for subsequent analysis.
- **Signal Smoothing:** A moving average filter is implemented to stabilize the waveform visualization and prevent erratic triggers.
- **Classification Logic:** Emotional states are categorized as **Calm** (near baseline), **Normal** (conversational levels), or **Angry** (high-intensity peaks).

## 5. Conclusion

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This implementation successfully demonstrates the viability of low-latency emotion recognition through acoustic analysis. The project serves as a robust foundation for future human-computer interaction research and real-time monitoring applications.